

MULTIVARIABLE CALCULUS TEXT (SUPPLEMENT TO OPEN STAX TEXT)



Karla Westphal
Saddleback College

Multivariable Calculus Text (Supplement to Open Stax Text)

This text is disseminated via the Open Education Resource (OER) LibreTexts Project (<https://LibreTexts.org>) and like the thousands of other texts available within this powerful platform, it is freely available for reading, printing, and "consuming."

The LibreTexts mission is to bring together students, faculty, and scholars in a collaborative effort to provide an accessible, and comprehensive platform that empowers our community to develop, curate, adapt, and adopt openly licensed resources and technologies; through these efforts we can reduce the financial burden born from traditional educational resource costs, ensuring education is more accessible for students and communities worldwide.

Most, but not all, pages in the library have licenses that may allow individuals to make changes, save, and print this book. Carefully consult the applicable license(s) before pursuing such effects. Instructors can adopt existing LibreTexts texts or Remix them to quickly build course-specific resources to meet the needs of their students. Unlike traditional textbooks, LibreTexts' web based origins allow powerful integration of advanced features and new technologies to support learning.



LibreTexts is the adaptable, user-friendly non-profit open education resource platform that educators trust for creating, customizing, and sharing accessible, interactive textbooks, adaptive homework, and ancillary materials. We collaborate with individuals and organizations to champion open education initiatives, support institutional publishing programs, drive curriculum development projects, and more.

The LibreTexts libraries are Powered by [NICE CXone Expert](#) and was supported by the Department of Education Open Textbook Pilot Project, the California Education Learning Lab, the UC Davis Office of the Provost, the UC Davis Library, the California State University Affordable Learning Solutions Program, and Merlot. This material is based upon work supported by the National Science Foundation under Grant No. 1246120, 1525057, and 1413739.

Any opinions, findings, and conclusions or recommendations expressed in this material are those of the author(s) and do not necessarily reflect the views of the National Science Foundation nor the US Department of Education.

Have questions or comments? For information about adoptions or adaptations contact info@LibreTexts.org or visit our main website at <https://LibreTexts.org>.

This text was compiled on 12/18/2025

TABLE OF CONTENTS

[Licensing](#)

[Karla's Supplement for Open Stax, section 4.2](#)

- [4.2: Supplement to Open Stax: Limits and continuity of functions of more than one variable](#)

[Index](#)

[Glossary](#)

[Detailed Licensing](#)

[Detailed Licensing](#)

Licensing

A detailed breakdown of this resource's licensing can be found in [Back Matter/Detailed Licensing](#).

CHAPTER OVERVIEW

Karla's Supplement for Open Stax, section 4.2

4.2: Supplement to Open Stax: Limits and continuity of functions of more than one variable

Karla's Supplement for Open Stax, section 4.2 is shared under a [CC BY-NC 4.0](#) license and was authored, remixed, and/or curated by Karla Westphal, Saddleback College.

4.2: Supplement to Open Stax: Limits and continuity of functions of more than one variable

Note

A quick note on notation: throughout this section, P will represent an arbitrary point in n -space, so that in 2-space P is (x,y) and in 3-space P is (x,y,z) . (P_0) will represent a fixed, but unspecified, point in n -space; in 2-space (P_0) is the point $((x_0, y_0))$ and in 3-space, (P_0) is the point $((x_0, y_0, z_0))$.

Definitions of limits (informal and formal)

One of the key concepts in calculus is that of a limit, and any time we work with a new type of function, we need to explore how this concept applies in our new setting.

We are already familiar with limits for two types of functions:

- Real-valued functions of one variable:

Examples:

$$\lim_{x \rightarrow -2} x^2 = 4 \quad \lim_{x \rightarrow 0} \frac{\sin x}{x} = 1$$

- Vector-valued functions of one variable:

$$\lim_{t \rightarrow 0} \left\langle \frac{\sin t}{t}, t^2 + 5 \right\rangle = \left\langle 1, 5 \right\rangle$$

Both of these satisfy the informal definition:

Definition: Informal definition of a limit

The limit of a function at a point is the unique value (if it exists) that the output gets close to as the input gets close to the specified point.

That informal definition applies to limits of all types of functions. What's different now is that the input is a point in n -space, rather than a single number. Informally,

$$\lim_{P \rightarrow P_0} f(P) = L \quad \text{means: as } P \rightarrow P_0, f(P) \text{ gets close to } L$$

$$\lim_{(x,y) \rightarrow (-2,3)} (xy) = -6$$

This simply tells us that as (x,y) gets close to $((-2,3))$, the product (xy) gets close to (-6) .

The definition above is informal because the concept of "close" is vague: two things are "close" if the distance between them is "small", but what constitutes "small?"

The formal definition simply introduces two variables:

(ϵ) :

This is a positive variable establishing a "standard of closeness" or "tolerance" for the outputs. We consider the output to be "close" to the limit if the distance between them is smaller than (ϵ) . We use a variable rather than a constant so that we are free to make this variable as small as we like, so that we can make the output get *arbitrarily* close to the limit. When the output (as it is for real-valued functions of multiple variables) is a number, this amounts to $|f(P) - L| < \epsilon$. Note that (ϵ) must be positive, because it is establishing a maximum allowable distance (and distance cannot be negative) between the output $(f(P))$ and the limit, L .

(δ) :

This is a positive variable establishing a "standard of closeness" or "tolerance" for the inputs. We will consider the input P to be close to (P_0) if the distance between them is smaller than (δ) . We'll use the notation $d(P, P_0)$ to designate the distance between these points. Note that $d(P, P_0) = \sqrt{(x-x_0)^2 + (y-y_0)^2}$ for 2-space and $d(P, P_0) = \sqrt{(x-x_0)^2 + (y-y_0)^2 + (z-z_0)^2}$ for 3-space. An arbitrary input P is "close" to the specified input (P_0) if $d(P, P_0) < \delta$.

Geometrically:

- In $\mathbb{R}^2$, the set of all points that are "close" to (P_0) will be an open disk of radius (δ) , centered at (P_0) .
- In $\mathbb{R}^3$, the set of all points that are "close" to (P_0) will be an open sphere of radius (δ) , centered at (P_0) .

This allows us to define limits formally. In the case of real-valued functions of more than one variable:

Definition: Formal definition of a limit for real-valued functions of more than one variable

$$\lim_{P \rightarrow P_0} f(P) = L \quad \text{means}$$

for every $(\epsilon > 0)$, there exists a $(\delta > 0)$ such that:

$$\text{if } d(P, P_0) < \delta, \text{ then } |f(P) - L| < \epsilon$$

This is a very formal way of saying that we can force the output to be as close to L as we like by making the input sufficiently close to (P_0) .

Evaluating limits

In this class, we will almost never evaluate limits by directly applying this definition. Instead, we'll create a toolbox of techniques we can use to evaluate limits.

Continuous functions

Earlier we gave the example that $\lim_{(x,y) \rightarrow (-2,3)} (xy) = -6$.

While we didn't formally justify this, it hardly seems surprising: nothing "weird" happens with the function at or near $(-2,3)$ and so the limit (the z-value we expect based on nearby behavior) is exactly the same as the function value we get when we plug in $(-2,3)$.

Any time "nothing weird happens" so that "we get what we expect," we say the function is **continuous**, and we define continuous functions now, both informally and formally.

Definition: Continuity

Generic, informal definition: A function is continuous at a point in its domain if

$\underbrace{\text{what we expect}}_{\text{the limit as we approach that point}} = \underbrace{\text{what we get}}_{\text{the function value at that point}}$

Formal Definition for real-valued functions of one variable: $f(x)$ is continuous at an x-value a in its domain if

$\lim_{x \rightarrow a} f(x) = f(a)$

Formal Definition for vector-valued functions of one variable: $\vec{r}(t)$ is continuous at a t-value a in its domain if

$\lim_{t \rightarrow a} \vec{r}(t) = \vec{r}(a)$

Formal Definition for real-valued functions of more than one variable: $f(P)$ is continuous at a point P_0 in its domain if

$\lim_{P \rightarrow P_0} f(P) = f(P_0)$

To evaluate limits of continuous functions at points in their domain, we simply plug in the point and evaluate the function. This, of course, begs the question: how do we know what functions are continuous?

There are two important theorems we need to be able easily determine when a function is continuous at a point. (More complicated functions might require more complicated tools, but these rules will suffice for our examples this semester.)

Theorem [1](#)

"Coordinate-identifying functions" (those that simply identify one of the coordinates of the input point) are continuous. In 2-space, there are two coordinate-identifying functions: $f(x,y) = x$ and $g(x,y) = y$. In 3-space, there are 3 such functions. We'll recycle the names "f" and "g"; it should be clear from the context which space we are working in. $f(x,y,z) = x$, $g(x,y,z) = y$, $h(x,y,z) = z$.

It is relatively easy to prove that these coordinate functions are continuous at an arbitrary point in their domain (and thus continuous everywhere on their domains) by using the formal definition of a limit to prove, for example, that $\lim_{(x,y) \rightarrow (x_0, y_0)} x = x_0$.

Theorem [2](#)

Any function built by combining continuous functions using any combination of arithmetic and/or composition is also continuous on its domain.

Example [1](#)

Explain how we know $k(x,y) = xy$ is continuous on its domain.

Solution

This function is continuous because it is the product of the two coordinate-identifying functions in 2-space:

$k(x,y) = \underbrace{f(x,y)}_x \cdot \underbrace{g(x,y)}_y$

Example [2](#)

Explain how we know $F(x,y) = e^{xy}$ is continuous on its domain.

Solution

This is a composition of the continuous function $k(x,y)$ from the previous example with the exponential function: $j(z) = e^z$, which we know to be continuous from first semester calculus. (Note that $j(z)$ is a real-valued function of one variable.)

Even though k and j are different types of functions, it's possible to compose them because the output of k is a real number, which is a legitimate input to j . The composition path is illustrated below:

$(x,y) \rightarrow \text{plug into } k \rightarrow xy \rightarrow \text{plug into } j \rightarrow e^{xy}$

$F(x,y) = j(k(x,y))$

Exercise [1](#)

See if you can explain why $G(x,y,z) = \sin(xy^2z^3)$ is continuous on its domain, which is all of $\mathbb{R}^3$.

Answer

We can build G out of the following continuous functions:

$$\left[\begin{array}{l} \text{left. } f(x,y,z) = x \backslash \text{right. } g(x,y,z) = y \backslash \text{right. } h(x,y,z) = z \end{array} \right] \text{ coordinate-identifying functions} \left[\begin{array}{l} f_1(w) = w^2 \backslash \text{right. } f_2(w) = w^3 \backslash \text{right. } f_3(w) = \sin(w) \end{array} \right] \text{ continuous real-valued functions of one variable}$$

$$(x = f(x,y,z))$$
 is continuous since it is a coordinate-identifying function

$$(y^2 = f_1(g(x,y,z)))$$
 is the composition of the squaring function with a coordinate-identifying function

$$(z^3 = f_2(h(x,y,z)))$$
 is the composition of the cubing function with a coordinate-identifying function

$$(p(x,y,z) = xy^2z^3)$$
 is the product of the three continuous functions above

$$(\sin(xy^2z^3) = f_3(p(x,y,z)))$$
 is the composition of the sine function with this continuous product

Since G is continuous,

$$\lim_{(x,y,z) \rightarrow (\pi, 2, -1)} \sin(xy^2z^3) = \sin(\pi \cdot 2^2 \cdot (-1)^3) = \sin(-4\pi) = 0$$

Almost all the functions we'll work with this semester will be continuous on their domains; this means that the "interesting" limits we'll evaluate will be as we approach a point not in the domain of the function.

Algebraic Simplification

Consider the following limit: $\lim_{(x,y) \rightarrow (0,0)} \frac{xy}{x^2 + y^2}$

While $\frac{xy}{x^2 + y^2}$ is continuous on its domain, we cannot evaluate this limit by plugging in $(0,0)$, since $(0,0)$ is not in the domain. When we try to plug in $(0,0)$ we see that both the top and bottom go to 0, leading to an indeterminate form: $\frac{0}{0}$.

However, just as with functions of one variable, when it is the same troublesome factor that makes top and bottom go to 0, we can simply cancel that factor:

$$\lim_{(x,y) \rightarrow (0,0)} \frac{xy}{x^2 + y^2} = \lim_{(x,y) \rightarrow (0,0)} \frac{y}{x + y} = 0$$

If we let $f_1(x,y) = \frac{xy}{x^2 + y^2}$ and $f_2(x,y) = \frac{y}{x + y}$, we see that they are the same function everywhere except when $x=0$, when f_1 is not defined but f_2 is. This means that they are the same *arbitrarily close to the origin* (as long as both are defined) and so both functions lead us to expect the same value as we get closer to the origin—i.e., they have the same limit.

Substitutions

Now consider $\lim_{(x,y) \rightarrow (0,0)} \frac{\overbrace{e^{(x+y)} - 1}^{\rightarrow 0}}{\underbrace{\sin(x+y)}_{\rightarrow 0}}$

Here, we have an indeterminate form of the type $\frac{0}{0}$, but there is no algebra we can do that will allow us to cancel things.

You might be inclined to do what we normally do when we encounter an indeterminate form: apply L'Hospital's rule. There is a slight problem with this: we haven't yet learned how to take the derivative of a real-valued function of more than one variable.

Hmmm. You might be inclined to employ the strategy of procrastination. Surely, soon, you will learn how to take the derivative of this new sort of function. Then you'll be able to apply L'Hospital's rule and evaluate this limit.

There is a fatal flaw in this way of thinking. You would be correct in thinking that soon we will learn to take derivatives of real-valued functions of more than one variable. But you would be wrong to think that this will enable us to apply L'Hospital's rule. For when we learn how to take the derivative of such a function, we will learn that the derivative is a vector! We cannot divide vectors, so we cannot take the quotient of the derivatives, as L'Hospital's rule would have us do. L'Hospital's rule simply does not apply.



Caution

L'Hospital's Rule is only valid for real-valued functions of one variable.

We need a different strategy.

You might notice that the same expression, $x+y$, occurs in the top and bottom of our function. If we let $(u = x+y)$ we can rewrite the entire limit in terms of just one variable, u . It is easy to see that as $(x,y) \rightarrow (0,0)$, $u \rightarrow 0$. (More formally:

$\lim_{(x,y) \rightarrow (0,0)} u = \lim_{(x,y) \rightarrow (0,0)} (x+y) = 0 + 0 = 0$). Thus the limit becomes $\lim_{u \rightarrow 0} \frac{e^u - 1}{\sin u}$. We still get an indeterminate form, but now we get it when taking the limit of a real-valued function of just one variable, so L'Hospital's rule applies!

$$\lim_{u \rightarrow 0} \frac{e^u - 1}{\sin u} = \lim_{u \rightarrow 0} \frac{\overbrace{e^u - 1}^{\rightarrow 0}}{\underbrace{\sin u}_{\rightarrow 0}} = \lim_{u \rightarrow 0} \frac{e^u}{\cos u} = \frac{e^0}{\cos 0} = \frac{1}{1} = 1$$

The trick of using a substitution to turn the problem into a problem involving only one variable is valid for functions of 3 variables as well.

? Exercise

Evaluate $\lim_{(x,y,z) \rightarrow (1, 1, -2)} \frac{\tan(x+y+z)}{2x+2y+2z}$

Answer

$$\lim_{(x,y,z) \rightarrow (1, 1, -2)} \frac{\overbrace{\tan(x+y+z)}^{\rightarrow 0}}{\underbrace{2x+2y+2z}_{\rightarrow 0}} = \lim_{(x,y,z) \rightarrow (1,1,-2)} \frac{\tan(x+y+z)}{2(x+y+z)}$$

Let $(u = x+y+z)$

As $(x,y,z) \rightarrow (1,1,-2)$, $u \rightarrow 0$

$$\lim_{(x,y,z) \rightarrow (1, 1, -2)} \frac{\tan(x+y+z)}{2(x+y+z)} = \lim_{u \rightarrow 0} \frac{\tan u}{2u} = \lim_{u \rightarrow 0} \frac{\overbrace{\tan u}^{\rightarrow 0}}{\underbrace{2u}_{\rightarrow 0}} = \lim_{u \rightarrow 0} \frac{\sec^2 u}{2} = \frac{\sec^2 0}{2} = \frac{1^2}{2} = \frac{1}{2}$$

✓ Example $\lim_{(x,y) \rightarrow (0,0)} \frac{\ln(x^2 + y^2)}{\ln(x^2 + y^2)}$

Evaluate $\lim_{(x,y) \rightarrow (0,0)} \frac{\ln(x^2 + y^2)}{\ln(x^2 + y^2)}$.

Solution

$$\lim_{(x,y) \rightarrow (0,0)} \frac{\ln(x^2 + y^2)}{\ln(x^2 + y^2)} = \lim_{(x,y) \rightarrow (0,0)} \frac{\ln(x^2 + y^2)}{\ln(x^2 + y^2)}$$

This limit leads to an indeterminate form of the type $\frac{0}{0}$.

To use L'Hospital's rule, we'll need to do some algebra to convert to an indeterminate quotient, and we can do that either before or after we do our substitution, in this case letting $u = x^2 + y^2$.

Note that this expression is never negative; we will need to reflect this in the limit. It's not enough to note that as $(x,y) \rightarrow (0,0)$, $u \rightarrow 0$; we need to record that it approaches 0 *from the right*:

$$\lim_{(x,y) \rightarrow (0,0)} \frac{\ln(x^2 + y^2)}{\ln(x^2 + y^2)} = \lim_{u \rightarrow 0^+} \frac{\ln(u)}{\ln(u)}$$

This is important because $\ln(u)$ would be undefined if u were negative. Continuing to evaluate the limit:

$$\lim_{(x,y) \rightarrow (0,0)} \frac{\ln(x^2 + y^2)}{\ln(x^2 + y^2)} = \lim_{u \rightarrow 0^+} \frac{\ln(u)}{\ln(u)}$$

Do the substitution.

$$\lim_{u \rightarrow 0^+} \frac{\ln(u)}{\ln(u)} = \lim_{u \rightarrow 0^+} \frac{\ln(u)}{\ln(u)}$$

Rewrite as an indeterminate quotient: instead of multiplying by u , divide by its reciprocal.

$$\lim_{u \rightarrow 0^+} \frac{\ln(u)}{\ln(u)} = \lim_{u \rightarrow 0^+} \frac{1}{\frac{1}{\ln(u)}}$$

Apply L'Hospital's rule.

$$\lim_{u \rightarrow 0^+} \frac{1}{\frac{1}{\ln(u)}} = \lim_{u \rightarrow 0^+} \frac{1}{\frac{1}{\ln(u)}}$$

Rewrite using positive exponents.

$$\lim_{u \rightarrow 0^+} \frac{1}{\frac{1}{\ln(u)}} = \lim_{u \rightarrow 0^+} \ln(u)$$

Dividing by a fraction is the same as multiplying by its reciprocal.

$$\lim_{u \rightarrow 0^+} \ln(u)$$

Cancel a factor of u .

$$\lim_{u \rightarrow 0^+} \ln(u)$$

Using Polar or Spherical coordinates for limits at the origin

You might notice that, had we switched to polar coordinates, the function in the last example would have been a function of just one variable, r , since $x^2 + y^2 = r^2$. Further, the origin is the one point in the (x,y) -plane that is completely determined by just its r -value in polar coordinates; every other point requires both an r and a θ -value, but we are approaching the origin if r goes to 0, regardless of the value of θ . Thus we could have rewritten the limit by switching to polar, with the choice that we will choose r to be non-negative. $\lim_{(x,y) \rightarrow (0,0)} \frac{\ln(x^2 + y^2)}{\ln(x^2 + y^2)} = \lim_{r \rightarrow 0^+} \frac{\ln(r^2)}{\ln(r^2)}$

You'll notice that, while r is allowed to take on negative values in polar coordinates, I stated that we would make a choice to restrict r to take on non-negative values only. There are two reasons that this is necessary:

1. If we allow r to take on both positive and negative values, we would introduce a discontinuity to a function that isn't present when we describe the function using rectangular coordinates. When working with limits, any substitution we do must preserve continuity.
2. While the expression $\sqrt{x^2 + y^2}$ does not appear in this particular example, it may very well occur in other examples. This expression is equal to r if r is not negative, but it is the opposite of r if r is negative.

For these reasons, any time we switch to polar coordinates, we will specify that we are switching to polar, *with the restriction that r is non-negative*. (It would also be correct to restrict r to taking on only non-positive values.)

✓ Example $\lim_{(x,y) \rightarrow (0,0)} \frac{\sin^2(\sqrt{x^2 + y^2})}{x^2 + y^2}$

Evaluate $\lim_{(x,y) \rightarrow (0,0)} \frac{\sin^2(\sqrt{x^2 + y^2})}{x^2 + y^2}$.

Solution

Switch to polar, with $r \geq 0$.

$$\lim_{(x,y) \rightarrow (0,0)} \frac{\sin^2(\sqrt{x^2 + y^2})}{x^2 + y^2} = \lim_{r \rightarrow 0^+} \frac{\sin^2(r)}{r^2}$$

This step would not be legal if we had not specified that $r \geq 0$ for the reasons outlined above.

$$\lim_{r \rightarrow 0^+} \frac{\sin^2(r)}{r^2} = \lim_{r \rightarrow 0^+} \frac{\sin^2(r)}{r^2}$$

Apply L'Hospital's rule

$$\lim_{r \rightarrow 0^+} \frac{\sin^2(r)}{r^2} = \lim_{r \rightarrow 0^+} \frac{2\sin(r)\cos(r)}{2r} = \lim_{r \rightarrow 0^+} \frac{\sin(2r)}{r}$$

We can use the product property of limits to separate the indeterminate form from the other factors.

$$\lim_{r \rightarrow 0^+} \frac{\sin(2r)}{r} = \lim_{r \rightarrow 0^+} \sin(2r) \cdot \lim_{r \rightarrow 0^+} \frac{1}{r}$$

Apply L'Hospital's rule again; evaluate the second limit by plugging in 0.

$$\lim_{r \rightarrow 0^+} \frac{\sin(2r)}{r} = \lim_{r \rightarrow 0^+} \frac{2\cos(2r)}{1} = 2$$

$$\lim_{r \rightarrow 0^+} \frac{\sin(2r)}{r} = 2$$

So far, switching to polar hasn't really introduced a new technique; it's just been a different substitution.

But switching to polar can be useful when we are approaching the origin, even if doing so requires us to use both variables: r and θ . This is illustrated in the next example.

✓ Example [\\(\PageIndex{5}\\)](#)

Evaluate $\lim_{(x,y) \rightarrow (0,0)} \frac{x^3}{\sqrt{x^2 + y^2}}$.

Solution

Switch to polar with $(r \geq 0)$.

$$\lim_{(x,y) \rightarrow (0,0)} \frac{x^3}{\sqrt{x^2 + y^2}} = \lim_{r \rightarrow 0^+} \frac{(r \cos(\theta))^3}{r}$$

Note that we can indicate that we are approaching the origin using just the variable r , even though both variables are needed to describe the function whose limit we are taking.

$$\lim_{r \rightarrow 0^+} \frac{r^3 \cos^3(\theta)}{r}$$

$$\lim_{r \rightarrow 0^+} \left(r^2 \cos^3(\theta) \right)$$

Cancel a factor of r .

It is tempting at this point to conclude that the limit is 0, since the first factor, (r^2) , clearly goes to 0. However, that fact by itself is not sufficient for us to conclude that the limit is 0. If the second factor were $(\tan(\theta))$ instead of $(\cos^3(\theta))$, this limit would not exist, since $(\tan(\theta))$ can go off to $(\pm \infty)$ so that we do not know that r (which is *going to, but not necessarily equal to* 0) will be small enough to make the product 0. It is necessary to point out that our second factor, $(\cos^3(\theta))$, is **bounded**. This simply means it does not go off to (∞) or $(-\infty)$, in this case, because it is always between (-1) and 1. To finish evaluating the limit, we will write

$$\lim_{r \rightarrow 0^+} \underbrace{r^2}_{\rightarrow 0} \underbrace{\cos^3(\theta)}_{\text{bounded}} = 0$$

In polar coordinates, if we are approaching the origin, we can describe this using just one variable: $(x,y) \rightarrow (0,0)$ is equivalent to $(r \rightarrow 0^+)$. Similarly, in spherical coordinates, we can indicate that we are approaching the origin by saying $(\rho \rightarrow 0^+)$. Here, (ρ) is positive by definition; not only must the limit be one-sided, it must be from the positive side.

? Exercise [\\(\PageIndex{3}\\)](#)

Evaluate $\lim_{(x,y,z) \rightarrow (0,0,0)} \frac{z^2}{\sqrt{x^2 + y^2 + z^2}}$

Answer

$$\lim_{(x,y,z) \rightarrow (0,0,0)} \frac{z^2}{\sqrt{x^2 + y^2 + z^2}} = \lim_{\rho \rightarrow 0^+} \frac{(\rho \cos(\phi))^2}{\rho}$$

Note that (ρ) is positive by definition; it is necessary to acknowledge this by taking a one-sided limit, but it is not necessary to state that $(\rho \geq 0)$ since this is a fact, not a choice we are making.

$$\lim_{\rho \rightarrow 0^+} \frac{\rho^2 \cos^2(\phi)}{\rho}$$

$$\lim_{\rho \rightarrow 0^+} \underbrace{\rho}_{\rightarrow 0} \underbrace{\cos^2(\phi)}_{\text{bounded}}$$

Cancel a factor of (ρ)

$$\lim_{\rho \rightarrow 0^+} \cos^2(\phi) = 0$$

✓ Example [\\(\PageIndex{6}\\)](#)

Evaluate $\lim_{(x,y) \rightarrow (0,0)} \frac{xy}{x^2 + y^2}$

Solution

Switch to polar with $(r \geq 0)$

$$\lim_{(x,y) \rightarrow (0,0)} \frac{xy}{x^2 + y^2} = \lim_{r \rightarrow 0^+} \frac{(r \cos(\theta))(r \sin(\theta))}{r^2} = \lim_{r \rightarrow 0^+} \frac{\cancel{r^2} \cos(\theta) \sin(\theta)}{\cancel{r^2}} = \lim_{r \rightarrow 0^+} \cos(\theta) \sin(\theta) \quad \& \text{ DNE}$$

This limit does not exist. Knowing that r is going to 0 tells us nothing about the value of (θ) . $(\cos(\theta) \sin(\theta))$ does not define a unique value.

⚠ Caution

If you switch to polar coordinates for limits approaching the origin, do not make any assumptions about the value that (θ) is approaching.

Similarly, if you are approaching the origin in 3-space, this means that $(\rho \rightarrow 0^+)$, but do not make any assumptions about the values of (θ) or (ϕ) .

Using Paths to show that the limit does not exist

Let's take a closer look at that last example. After canceling the factor of (r^2) , we were left with $(\cos(\theta) \sin(\theta))$.

We observed that r going to 0 puts no restrictions on (θ) . r going to 0 tells us we are approaching the origin; the value of θ tells us what direction we are approaching it from:

- If $(r \rightarrow 0^+)$ and $(\theta = 0)$, we are approaching the origin along the positive x -axis: We see that in this case $(\cos(\theta) \sin(\theta) = (1)(0) = 0)$.
- If $(r \rightarrow 0^+)$ and $(\theta = \frac{\pi}{2})$, we are approaching the origin along the positive y -axis. In this case $(\cos(\theta) \sin(\theta) = (0)(1) = 0)$.

- If $(r \rightarrow 0^+)$ and $(\theta = \frac{\pi}{4})$, we are approaching the origin along the portion of the line $y = x$ that lies in Quadrant I. In this case $(\cos \theta \sin \theta = \frac{1}{2})$ $\left(\frac{\sqrt{2}}{2} \right) \left(\frac{\sqrt{2}}{2} \right) = \frac{1}{2}$.

We get a different value when we approach from a different direction!!!

For real-valued functions of one variable, the concept of one-sided limits takes into account the direction from which we approach a point.

The multi-variable generalization of one-sided limits is limits along a path through a point.

Definition: Path limits

Define a **path** through a point (P_0) to be a smooth curve that passes through (P_0) . In order to take a limit of a function $f(P)$ along such a path, we require that the path is in the domain of f in some open region containing (P_0) , except possibly at (P_0) . If $(\vec{r}(t))$ is a smooth parametrization of such a curve, and $(\vec{r}(t))$ passes through (P_0) when $(t = t_0)$ $\lim_{t \rightarrow t_0} f(\vec{r}(t))$ $\text{along } C$ $f(x,y) = \lim_{t \rightarrow t_0} f(\vec{r}(t))$

Theorem $(\text{PageIndex}{3})$

If (C_1) and (C_2) are paths through (P_0) such that $\lim_{t \rightarrow t_0} f(\vec{r}_1(t)) \text{ along } C_1 \neq \lim_{t \rightarrow t_0} f(\vec{r}_2(t)) \text{ along } C_2$ or such that either of these limits does not exist, then $\lim_{t \rightarrow t_0} f(\vec{r}(t))$ does not exist.

Example $(\text{PageIndex}{7})$

$\lim_{(x,y) \rightarrow (0,0)} \frac{xy}{x^2 + y^2}$

Solution

We've already shown this limit does not exist by switching to polar coordinates. Here's what it looks like if we try paths.

The simplest paths through the origin are the x and y axes.

- We can parametrize the x -axis as $(\vec{r}_1(t) = \begin{pmatrix} t \\ 0 \end{pmatrix})$, noting that this path will pass through the origin when $t = 0$. (Note: this path corresponds to the rays we get in polar coordinates if $(r \geq 0)$ and (θ) is any multiple of (π) .)

$$\lim_{t \rightarrow 0} \frac{xy}{x^2 + y^2} \text{ along } \vec{r}_1(t) = \frac{t \cdot 0}{t^2 + 0^2} = 0$$

- Similarly, we can parametrize the y -axis as $(\vec{r}_2(t) = \begin{pmatrix} 0 \\ t \end{pmatrix})$; this path passes through the origin when $t = 0$.

$$\lim_{t \rightarrow 0} \frac{xy}{x^2 + y^2} \text{ along } \vec{r}_2(t) = \frac{0 \cdot t}{0^2 + t^2} = 0$$

Getting the same limit along 2 paths does not prove that this limit exists; we have checked two out of infinitely many possible paths. Noting that the line $y = x$ also passes through the origin (as previously noted, the portion of this line in Quadrant I is the ray $(\theta = \frac{\pi}{4}; r \geq 0)$), we parametrize it as $(\vec{r}_3(t) = \begin{pmatrix} t \\ t \end{pmatrix})$, noting that it also passes through the origin when $t = 0$.

$$\lim_{t \rightarrow 0} \frac{xy}{x^2 + y^2} \text{ along } \vec{r}_3(t) = \frac{t \cdot t}{t^2 + t^2} = \frac{1}{2}$$

Since 0 (the limit along paths 1 and 2) does not equal $(\frac{1}{2})$ (the limit along path 3), the overall limit does not exist.

Caution

- Paths can never be used to prove a limit exists, because it is never possible to check all infinitely many possible paths. If you keep getting the same limit, do one of two things:
 - Try a more complicated path (perhaps a non-linear path).
 - Run through all the techniques we've covered in this section, since all the others can be used both to show a limit does not exist and to find the limit when it does exist; perhaps there was a viable option that you rejected too quickly.

Unlike the technique of switching to polar coordinates, path limits can be used to show a limit does not exist at points other than the origin.

Example $(\text{PageIndex}{8})$

Evaluate $\lim_{(x,y) \rightarrow (1,-2)} \frac{2x+y}{y+2}$

Solution

- We can't plug in $(1,-2)$, since that makes both the top and bottom go to 0.
- The expression does not simplify.
- We do not have the same variable expression showing up on both the top and bottom, so no substitution looks promising.
- We are not approaching the origin, so switching to polar won't help.

Since none of the techniques we've learned for finding limits will work, let's hope the limit does not exist and try paths.

For limits in 2-space, there are three linear paths that are generally easy to use:

1. The line through the point that is parallel to the x -axis. This means the y -value must be constant; to pass through the point $(1,-2)$ it must be constant at (-2) : $(\vec{r}_1(t) = \begin{pmatrix} t \\ -2 \end{pmatrix})$ Note that this path will approach $(1,-2)$ as $(t \rightarrow 1)$. For this particular problem, we cannot use this path, as it lies entirely outside the domain of the function; if $y = -2$, the function is undefined because its denominator is 0.
2. The line through the point that is parallel to the y -axis. This means the x -value must be constant: $(\vec{r}_2(t) = \begin{pmatrix} 1 \\ t \end{pmatrix})$

Note that this path will approach $(1,-2)$ as $(t \rightarrow -2)$. For all other values of t , the function is defined, so this path is a good choice for this problem.

3. The line through the origin that also passes through $((1,-2))$. Using the origin as our "starting point" and $\langle \begin{matrix} 1, -2 \end{matrix} \rangle$ (the vector pointing from the origin to $((1,-2))$) as our direction vector, we get $\langle \begin{matrix} x \\ y \end{matrix} \rangle = \langle \begin{matrix} 0 \\ 0 \end{matrix} \rangle + t \langle \begin{matrix} 1, -2 \end{matrix} \rangle = \langle \begin{matrix} t \\ -2t \end{matrix} \rangle$. This path approaches $((1,-2))$ as $t \rightarrow 1$ and is in the domain of the function for all other t -values; it is a good choice as well.

We'll use the second and third options.
$$\lim_{(x,y) \rightarrow (1,2)} \frac{\text{along } \vec{r_1(t)}}{\frac{2x+y}{y+2}} \quad \&\& \quad \lim_{t \rightarrow -2} \frac{2(1+t)}{t+2} \quad \&\& \quad \lim_{t \rightarrow -2} \frac{t+2}{t+2} \quad \&\& \quad 1$$

$$\begin{aligned} \lim_{\substack{\text{stackrel{(x,y)}{\rightarrow}(1,2)}{\text{along}}}} \frac{\vec{r_2(t)}}{\frac{2x+y}{y+2}} \quad \& \lim_{\substack{\rightarrow 1}} \frac{2t-2t}{-2t+2} \quad \& \lim_{\substack{\rightarrow 1}} \frac{0}{-2t+2} \quad \& \& \end{aligned} \quad \nonumber$$

$(1 \neq 0)$, so $\lim_{(x,y) \rightarrow (1,-2)} \frac{2x+y}{y+2}$ does not exist.

 Note

Easy paths to try through $((x_0, y_0))$ in 2-space:

- The path through the point parallel to the x-axis: $\langle \vec{r}_1(t) = \langle \begin{matrix} t, y_0 \end{matrix} \rangle \rangle$
- The path through the point parallel to the y-axis: $\langle \vec{r}_2(t) = \langle \begin{matrix} x_0, t \end{matrix} \rangle \rangle$
- For points other than the origin, the line through the origin and the point: $\langle \vec{r}_3(t) = \langle \begin{matrix} x_0 t, y_0 t \end{matrix} \rangle \rangle$. (If the point $\langle (x_0, y_0) \rangle$ is the origin, any linear path of the form $\langle \begin{matrix} at, bt \end{matrix} \rangle$ for real numbers a and b , not both 0, is a possible choice.)

Easy paths through $((x_0, y_0, z_0))$ in 3-space:

- The path through the point parallel to the x-axis: $\vec{r}_1(t) = \begin{pmatrix} t \\ y_0 \\ z_0 \end{pmatrix}$.
- The path through the point parallel to the y-axis: $\vec{r}_2(t) = \begin{pmatrix} x_0 \\ t \\ z_0 \end{pmatrix}$.
- The path through the point parallel to the z-axis: $\vec{r}_3(t) = \begin{pmatrix} x_0 \\ y_0 \\ t \end{pmatrix}$.
- For points other than the origin, the line through the origin and the point: $\vec{r}_4(t) = \begin{pmatrix} x_0 t \\ y_0 t \\ z_0 t \end{pmatrix}$. (If the point $\vec{r}(x_0, y_0, z_0)$ is the origin, any linear path through the origin of the form $\vec{r}(t) = \begin{pmatrix} at \\ bt \\ ct \end{pmatrix}$ for real numbers a, b, and c, not all zero, is a possible choice.

? Exercise \(\backslash\PageIndex{4}\)

Evaluate. $\lim_{(x,y,z) \rightarrow (-2, 1, -1)} \frac{x+y-z}{z+1}$

Answer

Note that since $\sqrt{z+1}$ appears in the denominator, we cannot use any path for which z is constant. We can, however, try the following two paths:

Path 1 (parallel to the z-axis): $\vec{r}_1(t) = \begin{pmatrix} -2 \\ 1 \\ t \end{pmatrix}$

$$\begin{aligned} \lim_{\stackrel{\text{along}}{\rightarrow}} \frac{\text{vec}_r(1)(t)}{\text{frac}{x+y-z}{z+1}} &= \lim_{\rightarrow -1} \frac{-2 + 1 - t}{t+1} \parallel \lim_{\rightarrow -1} \frac{-1-t}{t+1} \parallel \lim_{\rightarrow -1} \frac{(-1)(1+t)}{t+1} \parallel \& -1 \end{aligned}$$

Path 2 (line through origin and $\mathbf{(-2, 1, -1)}$): $\mathbf{r_2}(t) = \begin{pmatrix} -2t \\ t \\ -t \end{pmatrix}$

$$\begin{aligned} \lim_{\text{stackrel{\scriptstyle}{(x,y,z)} \rightarrow (-2, 1, -1)}} \frac{\text{vec}\{r_2(t)\}}{x+y-z} &= \lim_{t \rightarrow 1} \frac{-2t + t - (-t)}{-t+1} = \\ \lim_{t \rightarrow -1} \frac{0}{-t+1} &= 0 \end{aligned}$$

Since $\lim_{(x,y,z) \rightarrow (-2, 1, -1)} \frac{x+y+z}{z+1}$ does not exist.

In the examples so far, we've been able to use only linear paths. As a general rule, these tend to be simpler than non-linear paths, so it definitely makes sense to start with them. However, a path through a point can be any smooth curve that passes through a point: it need not be a straight line. In the example below, we show that it is sometimes necessary to use a non-linear path.

✓ Example \(\backslash PageIndex{9}\)

$$\lim_{(x,y) \rightarrow (0,0)} \frac{x^2 y}{x^4 + y^4}$$

Solution

- Path 1 (the x-axis): $\vec{r}_1(t) = \begin{pmatrix} t \\ 0 \end{pmatrix}$

$$\lim_{(x,y)\rightarrow(0,0)}\frac{x^2y}{x^4+y^4}=\lim_{t\rightarrow 0}\frac{\left(\frac{t^2}{2}\right)(0)}{t^4+0^4}=\lim_{t\rightarrow 0}\frac{0}{t^4}=\frac{0}{0}$$

- Path 2 (the y-axis): $\vec{r}_2(t) = \begin{pmatrix} 0, t \end{pmatrix}$

$$\lim_{\stackrel{\text{along } \vec{r_2(t)}}{\rightarrow}(0,0)} \frac{x^2y}{x^4 + y^2} = \lim_{\rightarrow 0} \frac{\left(0^2\right) t}{0^4 + t^2} = \lim_{\rightarrow 0} \frac{0}{t^2} = 0$$

The fact that we got the same limit along two (out of infinitely many possible!) paths does NOT tell us the limit is 0. We need to keep going.

Next, we show that using any other line through the origin leads to the same limit of 0. "Path 3" is really a family of infinitely many paths.

- "Path 3" (a line of slope $(m \neq 0)$ through the origin): $(\vec{r}_3)(t) = \begin{pmatrix} t \\ mt \end{pmatrix}$

$$\begin{aligned} & \lim_{(x,y) \rightarrow (0,0)} \frac{\vec{r}_3(t)}{x^2 y} \{ \frac{x^4 + y^4}{(t^2 \text{right})(mt)} \{ t^4 + (mt)^2 \} \} \\ & \lim_{t \rightarrow 0} \frac{\overbrace{mt^3}^{\text{right}}}{\underbrace{t^4 + m^2 t^2}_{\text{right}}} \} \} \underbrace{\quad}_{\text{LH}} \} \lim_{t \rightarrow 0} \frac{\overbrace{3mt^2}^{\text{right}}}{\underbrace{4t^3 + 2m^2 t}_{\text{right}}} \} \} \underbrace{\quad}_{\text{LH}} \} \lim_{t \rightarrow 0} \frac{6mt}{12t^2 + 2m^2} \} \} \\ & \lim_{t \rightarrow 0} \frac{6m(0)}{12\text{left}(0^2\text{right}) + 2m^2} \} \} \lim_{t \rightarrow 0} \frac{0}{2m^2} \} \} \end{aligned}$$

Finally, let's try a parabolic path through the origin.

- Path 4 (the parabola $(y = x^2)$): $(\vec{r}_4)(t) = \left\langle \begin{matrix} t \\ t^2 \end{matrix} \right\rangle$

$$\begin{aligned} \lim_{\substack{\rightarrow \\ (x,y) \rightarrow (0,0)}} \frac{\text{t{x}{along} \vecc{r_4}(t)}}{\frac{\text{f}{x^2y}}{x^4+y^2}} &= \lim_{\rightarrow 0} \frac{\text{f}{t^2 \left\langle t^2 \right\rangle}}{t^4 + \left\langle t^2 \right\rangle^2} \\ &= \lim_{\rightarrow 0} \frac{\text{f}{t^4}}{t^4 + t^4} = \lim_{\rightarrow 0} \frac{\text{f}{t^4}}{2t^4} = \frac{\text{f}{1}}{2} \end{aligned}$$

$\frac{1}{2} \neq 0$ so $\lim_{(x,y) \rightarrow (0,0)} \frac{x^2 y}{x^4 + y^2}$ does not exist.

4.2: Supplement to Open Stax: Limits and continuity of functions of more than one variable is shared under a [CC BY-NC 4.0](#) license and was authored, remixed, and/or curated by Karla Westphal, Saddleback College.

Glossary

Sample Word 1 | Sample Definition 1

Detailed Licensing

Overview

Title: [Multivariable Calculus Text \(Supplement to Open Stax Text\)](#)

Webpages: 13

Applicable Restrictions: Noncommercial

All licenses found:

- [CC BY-NC 4.0](#): 92.3% (12 pages)
- [Undeclared](#): 7.7% (1 page)

By Page

- [Multivariable Calculus Text \(Supplement to Open Stax Text\)](#)
 - [CC BY-NC 4.0](#)
 - [Front Matter](#) - [CC BY-NC 4.0](#)
 - [TitlePage](#) - [CC BY-NC 4.0](#)
 - [InfoPage](#) - [CC BY-NC 4.0](#)
 - [Table of Contents](#) - [CC BY-NC 4.0](#)
 - [Licensing](#) - [CC BY-NC 4.0](#)
 - [Karla's Supplement for Open Stax, section 4.2](#) - [CC BY-NC 4.0](#)
 - [4.2: Supplement to Open Stax: Limits and continuity of functions of more than one variable](#) - [CC BY-NC 4.0](#)
 - [Back Matter](#) - [CC BY-NC 4.0](#)
 - [Index](#) - [CC BY-NC 4.0](#)
 - [Glossary](#) - [CC BY-NC 4.0](#)
 - [Detailed Licensing](#) - [CC BY-NC 4.0](#)
 - [Detailed Licensing](#) - [Undeclared](#)

Detailed Licensing

Overview

Title: [Multivariable Calculus Text \(Supplement to Open Stax Text\)](#)

Webpages: 13

Applicable Restrictions: Noncommercial

All licenses found:

- [CC BY-NC 4.0](#): 92.3% (12 pages)
- [Undeclared](#): 7.7% (1 page)

By Page

- [Multivariable Calculus Text \(Supplement to Open Stax Text\)](#)
 - [CC BY-NC 4.0](#)
 - [Front Matter](#) - [CC BY-NC 4.0](#)
 - [TitlePage](#) - [CC BY-NC 4.0](#)
 - [InfoPage](#) - [CC BY-NC 4.0](#)
 - [Table of Contents](#) - [CC BY-NC 4.0](#)
 - [Licensing](#) - [CC BY-NC 4.0](#)
 - [Karla's Supplement for Open Stax, section 4.2](#) - [CC BY-NC 4.0](#)
 - [4.2: Supplement to Open Stax: Limits and continuity of functions of more than one variable](#) - [CC BY-NC 4.0](#)
 - [Back Matter](#) - [CC BY-NC 4.0](#)
 - [Index](#) - [CC BY-NC 4.0](#)
 - [Glossary](#) - [CC BY-NC 4.0](#)
 - [Detailed Licensing](#) - [CC BY-NC 4.0](#)
 - [Detailed Licensing](#) - [Undeclared](#)